

Summary of Research Protocol Submitted for Ethical Clearance
Cover Letter to the Ethics and Research Committee
OAUTHC, Ile-Ife | 16 August 2026

Submission Summary

Full title: External Validation of the Fuzzy Audiometric Index (FAI), a Fuzzy Inference System for Automated Audiogram Classification, Using Clinical Audiometry Data from the Department of Otorhinolaryngology, Obafemi Awolowo University Teaching Hospitals Complex (OAUTHC), Ile-Ife

Investigators: Dr Sanyaolu Alani Ameye, MBBS, FWACS, FMCORL (Principal Investigator, former Reader, Department of Otorhinolaryngology, OAU) and Dr Samuel Ayodele, Consultant ENT Surgeon (Co-Investigator and Site Lead, OAUTHC)

What the Study Does

The study externally validates the Fuzzy Audiometric Index, an interpretable fuzzy inference system that classifies audiograms into WHO hearing-loss severity grades. The FAI was developed on United States survey data (NHANES, 19,568 ears) where it achieved a weighted kappa of 0.931 against the WHO PTA-4 reference, with 94.7% overall agreement and 79.8% agreement in borderline cases. It has never been tested on West African clinical audiometry, and this study provides that evidence.

Study Design

- **Design:** Retrospective chart review of archived audiometry records, 1 January 2020 to 30 June 2026. No patient contact, no intervention, no prospective data collection.
- **Sample:** Minimum 400 ears from approximately 200 to 250 patients (minimum 250 ears for the study to proceed), drawn consecutively from the ENT audiology records.
- **Reference standards:** WHO PTA-4 severity grade (primary) and documented consultant ENT clinical grading where available (secondary).
- **Analysis:** Weighted Cohen's kappa, overall accuracy, borderline-zone performance (within 5 dB of a severity boundary), configuration classification, age/sex/aetiology subgroup comparisons, and continuous score agreement (correlation, mean absolute error, Bland-Altman). Reported per the TRIPOD statement for external validation.

- **Key methodological feature:** The FAI is frozen; no re-fitting on OAUTHC data. This is a genuine external validation, not a re-development.

Ethics and Data Governance

- **Consent:** Waiver of consent is requested. The study is retrospective, uses only existing records, and involves no more than minimal risk.
- **Anonymity:** All data are de-identified at extraction (k-anonymity, groups of at least 5). No names, hospital numbers, addresses, or other identifiers leave the hospital.
- **Data security:** Data are stored on encrypted, password-protected media. Aggregated, fully de-identified analysis files only are shared with the principal investigator outside the hospital.
- **Compliance:** The protocol follows NHREC requirements for retrospective research using archived records.

What We Request

Approval of the protocol, including the waiver of informed consent, to enable the Department of Otorhinolaryngology to extract and analyse de-identified archived audiometry data within the study window.

Attachments: Full research protocol, data dictionary, extraction template, investigator CVs.

Principal Investigator | Dr Sanyaolu Alani Ameye, MBBS, FWACS, FMCORL **Site Lead** | Dr Samuel Ayodele, Consultant ENT Surgeon, OAUTHC